

QUICK SORT

CODE:

```
✓ #include <stdio.h>
  #include <stdlib.h>
  #include <time.h>
✓ void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
  }
✓ int hoarePartition(int arr[], int low, int high) {
    int pivot = arr[low];
    int i = low - 1;
    int j = high + 1;
    while (1) {
      do {
        i++;
      } while (arr[i] < pivot);
      do {
        j--;
      } while (arr[j] > pivot);
      if (i >= j)
        return j;
      swap(&arr[i], &arr[j]);
    }
  }
✓ void quickSort(int arr[], int low, int high) {
  if (low < high) {
    int p = hoarePartition(arr, low, high);
    quickSort(arr, low, p);
    quickSort(arr, p + 1, high);
  }
}
```

```

void choosePivot(int arr[], int low, int high, int choice) {
    int pivotIndex;
    if (choice == 1) {
        return;
    }
    else if (choice == 2) {
        swap(&arr[low], &arr[high]);
    }
    else if (choice == 3) {
        pivotIndex = low + rand() % (high - low + 1);
        swap(&arr[low], &arr[pivotIndex]);
    }
}

void printArray(int arr[], int n) {
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    printf("\n");
}

int main() {
    int arr[] = {157,110,147,122,111,149,151,141,123,112,117,133};
    int n = sizeof(arr) / sizeof(arr[0]);
    int choice;
    srand(time(NULL));
    printf("Choose Pivot:\n");
    printf("1. First Element\n");
    printf("2. Last Element\n");
    printf("3. Random Element\n");
    printf("Enter choice: ");
    scanf("%d", &choice);
    choosePivot(arr, 0, n - 1, choice);
    quickSort(arr, 0, n - 1);

    printf("Sorted Array:\n");
    printArray(arr, n);
    return 0;
}

```

OUTPUT:

```

PS C:\Users\chetan kumar g\OneDrive - Amrita Vishwa Vidyapeetham- Chennai Campus\IV SEM\Principles of Functional Languages\class practice\23 Jan> ./1
Choose Pivot:
1. First Element
2. Last Element
3. Random Element
Enter choice: 1
Sorted Array:
110 111 112 117 122 123 133 141 147 149 151 157
PS C:\Users\chetan kumar g\OneDrive - Amrita Vishwa Vidyapeetham- Chennai Campus\IV SEM\Principles of Functional Languages\class practice\23 Jan> ./1
Choose Pivot:
1. First Element
2. Last Element
3. Random Element
Enter choice: 2
Sorted Array:
110 111 112 117 122 123 133 141 147 149 151 157
PS C:\Users\chetan kumar g\OneDrive - Amrita Vishwa Vidyapeetham- Chennai Campus\IV SEM\Principles of Functional Languages\class practice\23 Jan> ./1
Choose Pivot:
1. First Element
2. Last Element
3. Random Element
Enter choice: 3
Sorted Array:
110 111 112 117 122 123 133 141 147 149 151 157

```

COUNCLUSION:

Random pivot is the best in comparison with first and last

So according to my observation

Random pivot >>> First pivot >> Last pivot

First pivot

157 | 110 | 147 | 122 | 111 | 149 | 151 | 141 | 123 | 112 | 117 | 134

Step 1

133 | 147 | 122 | 111 | 149 | 151 | 141 | 123 | 112 | 117 | 157

Step 2

123 | 117 | 122 | 111 | 112 | 123 | 141 | 151 | 149 | 147 | 157

Step 3

112 | 117 | 122 | 111 | 123 | 133 | 141 | 151 | 149 | 147 | 157

Step 4

117 | 112 | 122 | 117 | 123 | 133 | 141 | 151 | 149 | 147 | 157

Step 5

111 | 112 | 122 | 117 | 123 | 133 | 141 | 151 | 144 | 147 | 157

Step 6

111 | 112 | 117 | 122 | 123 | 133 | 141 | 141 | 131 | 144 | 147 | 157

Step 7

111 | 112 | 117 | 122 | 122 | 133 | 141 | 147 | 144 | 151 | 157

Step 8

111 | 112 | 117 | 122 | 123 | 133 | 141 | 144 | 147 | 151 | 157

157	110	147	122	111	149	151	141	123	112	117	133
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Step 1

117	110	112	122	111	123	133	151	141	149	147	157
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Step 2

117	110	112	122	111	123	133	151	141	149	147	157
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Step 3

110	111	112	122	117	123	133	151	141	149	147	157
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Step 4

110	111	112	117	122	123	133	151	141	149	147	157
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Step 5

110	111	112	117	122	123	133	151	141	149	147	157
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Step 6

110	111	112	117	122	123	133	141	147	149	151	157
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Step 7

110	111	112	117	122	123	133	141	147	149	151	157
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Random

157	110	147	122	111	149	151	141	123	112	117	133
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Step 1 111

111	110	147	122	157	149	151	141	123	112	117	133
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Step 2 111

110	111	147	122	157	149	151	141	123	112	117	133
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Step 3 151

110	111	147	122	133	149	117	141	123	112	151	137
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Step 4 122

110	111	112	117	122	149	133	141	123	147	151	137
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Step 5 133

110	111	112	117	122	123	133	141	149	147	151	137
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Step 6 147

110	111	112	117	122	123	133	141	149	147	151	137
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